

NEST-DDI: AUDITABLE TEMPORAL MODELING FOR PROSPECTIVE DRUG-DRUG INTERACTION REPORT MONITORING

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ABSTRACT

Drug-drug interaction (DDI) systems are commonly evaluated with random splits, although spontaneous reports are selective and time-stamped. We introduce a cutoff-safe benchmark that fixes candidate risk sets and feature availability before each quarter. It separates pair recurrence, first reporting of an unseen pair, and first reporting of an unseen pair-event combination. On release-local FAERS/AEMS data (2024Q3–2025Q3), NEST-DDI combines static pair background with multi-scale history. On the frozen 2025Q1→2025Q2 recurrence test, NEST-DDI obtains AUPR 0.90526 versus 0.90437 for histogram-based gradient boosting (HGB); the later 2025Q2→2025Q3 hold-out gives 0.91285 versus 0.91174. The recurrence gain is small, while a fixed rank fusion of NEST-DDI and GraphSAGE improves first-pair AUPR (0.16407) without broader utility. The contribution is an auditable prospective protocol and matched baseline suite for report triage; both are released at github.com/zhaoqian1314/nest-ddi.

Index Terms— drug-drug interaction, pharmacovigilance, temporal evaluation, FAERS, benchmark

1. INTRODUCTION

Spontaneous reports are not exposure-normalized observations: reporting depends on use, attention, regulation and data revision. Thus an unreported pair is not a confirmed negative, and a random split can leak reporting regimes across train and test. Disproportionality and data-driven interaction screening answer related but distinct retrospective questions, not whether a previously unseen pair will be reported next [1, 2, 3, 4].

We formulate prospective monitoring at cutoff c . Let H_c contain reports available by c and let the candidate universe be fixed before evaluation. We predict (i) whether a historical pair recurs in the next quarter, (ii) whether an unseen pair is first reported, and (iii) whether an unseen pair-MedDRA event triplet is first reported. This separation makes the estimate, risk set and negative construction explicit.

Our contributions are: (1) a release-local, receipt-time audit for the FDA quarterly report releases (FAERS, now

the Adverse Event Monitoring System, AEMS); (2) matched statistical, heuristic, tree, tabular-neural and historical-graph baselines; and (3) NEST-DDI, a structural-background plus multi-scale-history comparator, evaluated with a later temporal holdout and conservative claims. Drug normalization follows the DDInter inventory [5], while independent label-text checks use openFDA [6].

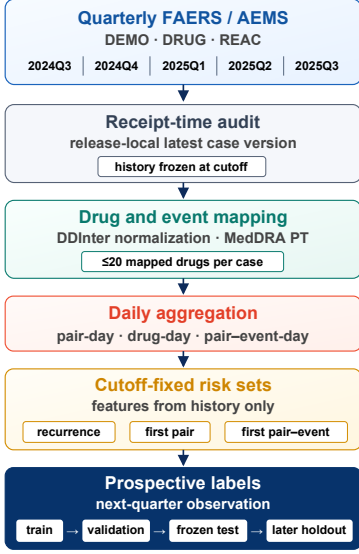
1.1. Why a benchmark is the contribution

The benchmark is not merely a new split of an existing DDI table. It defines the object being predicted, the information allowed to construct each feature, and the denominator against which a ranking is interpreted. These three contracts are essential in pharmacovigilance because the observation process changes over time and because absent reports mix true negatives, unobserved exposure and delayed ascertainment, creating a positive-unlabeled setting [7]. A model can therefore obtain a high random-split score while failing the operational question: which candidates should be reviewed before the next reporting period?

The protocol also makes methods comparable across input families. Molecular, multimodal and knowledge-graph systems such as DeepDDI, DPDDI, DDIMDL, Decagon, SumGNN, MRCGNN and TIGER use curated structures, relations or heterogeneous graphs [8, 9, 10, 11, 12, 13, 14]; temporal graph models such as TGAT and TGN assume event streams with their own negative-sampling contracts [15, 16]. Without a cutoff-fixed risk set, an apparent advantage may reflect future-derived vocabulary, case-version replacement, or a different candidate denominator rather than a better ranker. Our benchmark supplies a common 41-feature contract for matched tabular and historical-graph comparisons, while explicitly marking molecular/KG models as data-mismatched rather than silently adapting them.

Finally, the benchmark reports both positive and negative evidence. It records small recurrence gains, saturated top- K precision, sampling sensitivity, role-stratified differences and the absence of a later fusion gain. This prevents a single leaderboard number from being mistaken for causal interaction strength or clinical incidence. The resulting artifact is useful for method selection, audit and preregistration: every

A Cutoff-safe dataset construction



B NEST-DDI computational graph

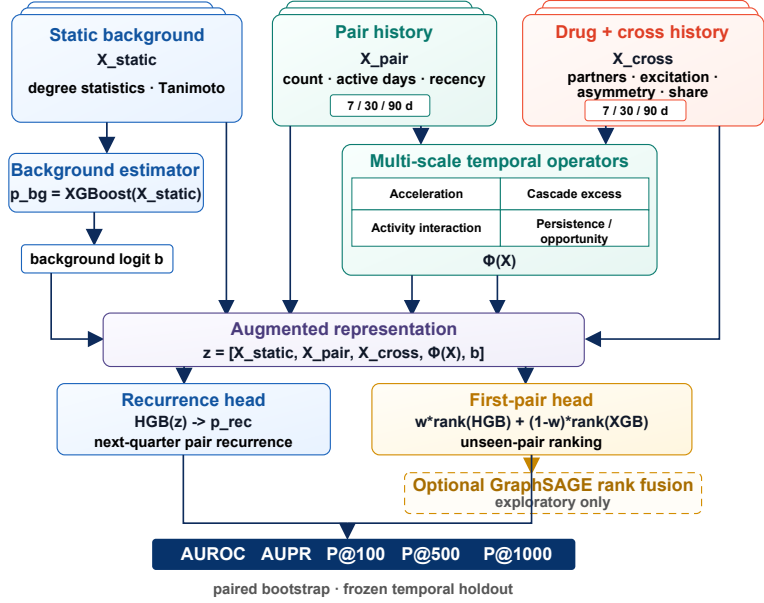


Fig. 1. Implementation-grounded framework. Left: receipt-time-safe construction of rolling FAERS/AEMS prediction panels. Right: NEST-DDI maps static, pair-history and drug-history inputs through deterministic temporal operators, injects a static-background logit, and applies task-specific ranking heads.

score can be traced to a release, cutoff, risk-set size, seed and label-opening date.

2. TEMPORAL BENCHMARK AND MODEL

We process quarterly FAERS/AEMS ASCII releases from 2024Q3 through 2025Q3 [1] (Fig. 1). Within each release, only the latest case version is retained; later releases cannot rewrite historical features. Drug names are normalized to the DDInter inventory, cases with more than 20 mapped drugs are excluded, and each case contributes unique unordered drug pairs and MedDRA preferred terms. The canonical four-quarter panel is distributed with the code; the release-local pair-day file contains 3,005,177 rows and 177,902 observed pairs.

For pair $p = (a, b)$, the recurrence label is $Y_{p,c}^{\text{rec}} = 1$ if p appears in $(c, c + 1]$. First-pair evaluation is restricted to p absent from H_c . For first pair–event discovery, the event vocabulary is selected from H_c only; eligible unseen triplets are sampled with a documented seed and denominator, so sampled AUPR is not a prevalence estimate.

NEST-DDI uses static graph degree and molecular similarity from versioned drug records [17] as a structural background, then adds pair and drug report count, active days, time since last report, and 7/30/90-day exponential decays. Geometric means, asymmetry and pair shares model cross-drug history. We compare recency/popularity heuristics [18], EdgeBank-style memory [19], logistic re-

gression, random forest, ExtraTrees, HGB, XGBoost [20], MLP, DeepDDI-style DNN, residual MLP, Wide&Deep, DeepFM, FT-Transformer, 1-D CNN, DPDDI-style GCN and GraphSAGE [21] history fusion. We also report prospective binary adaptations of DRIFT, DrugDAGT and SMR-DDI [22, 23, 24]; their original checkpoint and multi-class contracts differ from this benchmark. Neural models use the same frozen candidate rows; graph edges use co-medication observed before the cutoff only. Continuous-time point-process alternatives are acknowledged but not claimed as exact reproductions because Neural Hawkes and Transformer Hawkes require event-time likelihood contracts [25, 26].

3. EXPERIMENTAL PROTOCOL

Rolling cutoffs 2024Q3, 2024Q4 and 2025Q1 define next-quarter folds; 2025Q1→2025Q2 is the frozen test. A later holdout predicts 2025Q2→2025Q3 with choices frozen. We report AUROC, AUPR and precision@K ($K \in \{100, 500, 1000\}$). Paired uncertainty uses 300 stratified bootstrap resamples. For event discovery, 2024Q3→Q4 is training, Q4→2025Q1 validation and Q1→Q2 testing, with all positives and ten sampled negatives per positive.

Table 1. Frozen pair panel. AUPR is primary. † denotes a prospective binary adaptation; original checkpoint or multi-class label contracts differ.

Method	Rec-AUROC	Rec-AUPR	First-AUROC	First-AUPR
Recency–popularity	0.9762	0.89444	0.8991	0.13253
DRIFT-Temporal†	0.87124	0.57142	0.72649	0.04116
DrugDAGT-Temporal†	0.56954	0.15807	0.53672	0.01751
SMR-DDI-Temporal†	0.86529	0.55594	0.66384	0.03205
ExtraTrees	0.9776	0.90192	0.9018	0.13776
HGB	0.97810	0.90437	0.90384	0.14570
FT-Transformer	0.97831	0.90448	0.90568	0.14387
NEST-DDI	0.97832	0.90520	0.90407	0.14870

Table 2. Frozen first pair–event discovery metrics (10 sampled negatives per positive).

Method	AUROC	sAUPR	P@100	P@500	P@1000
XGB (val.)	0.79384	0.29806	0.720	0.650	0.621
HGB (full)	0.79272	0.29588	0.780	0.694	0.672

4. RESULTS

NEST-DDI improves recurrence AUPR over HGB by 0.00090 (95% CI 0.00062–0.00120) and AUROC by 0.00022 (Table 1). On the later holdout, AUPR is 0.91285 versus 0.91174 for HGB (difference 0.00110, 95% CI 0.00081–0.00139). These are statistically detectable but practically modest gains. For first-pair ranking, the fixed equal-rank fusion reaches AUPR 0.16407 and AUROC 0.90964; the triage-weighted version reaches P@100 0.40. These fusion results are exploratory and were not used to tune recurrence.

For first pair–event discovery, the frozen case–control test contains 7,466,682 eligible unseen triplets before sampling. Table 2 reports the held-out metrics with matched baselines alone, as NEST-DDI is not evaluated on this task; the differing AUPR and P@100 rankings illustrate why event-level discovery and pair recurrence should not share one leaderboard.

The paired bootstrap gains for fixed fusion over optimized NEST are +0.01468 AUPR (95% CI [0.01103, 0.01850]) and +0.00507 AUROC ([0.00413, 0.00605]); over GraphSAGE they are +0.01371 and +0.00661, respectively.

4.1. Dataset and risk-set diagnostics

Table 3 reports the cutoff-specific scale. The candidate universe is fixed from the DDInter inventory, whereas labels are opened only in the subsequent quarter. The ratio of future positives is therefore a property of the reporting process, not a user-selected class balance. In the first-pair task, the positive rate is lower because all historically observed pairs are removed from the candidate set.

Table 3. Release-local benchmark scale and target prevalence.

Cutoff	Cases	Pairs	Rec.+	First+
2024Q3	162,538	93,194	17,804	4,118
2024Q4	173,234	97,943	18,226	4,506
2025Q1	169,726	94,655	19,041	4,772
2025Q2	162,600	92,631	19,388	N/A

4.2. Baseline breadth and ablations

The recurrence leaderboard is history-dominated, but the relative ordering is not explained by one feature alone. Under the explicit feature contract in the machine-readable audit, static-only, history-only, no-decay, no-drug-analogue and full NEST-DDI variants obtain AUPR 0.55045, 0.90358, 0.90537, 0.90381 and 0.90526, respectively. Thus history features account for most of the signal, while the drug-analogue block contributes a small complementary increment. The no-decay variant is marginally above the full model on this frozen fold, so decay operators are not claimed as a universal improvement. The same ablation pattern is weaker for first-pair prediction (Fig. 3), where GraphSAGE contributes complementary ranking information. Feature lists, model parameters, the input hash and per-sample scores are released in the machine-readable ablation audit accompanying the code. We additionally evaluated score calibration and ranking stability, treating calibration as distinct from discrimination [27]. On the later holdout, NEST-DDI has Brier score 0.03301 and expected calibration error (ECE) 0.00346, compared with 0.03334 and 0.00688 for HGB. At top-list capacities of 100, 500 and 1,000, NEST-DDI precisions are 1.00, 1.00 and 1.00 on the frozen recurrence test; the corresponding HGB values are 1.00, 1.00 and 0.99. These saturated values motivate reporting AUPR in addition to precision@K.

4.3. Sampling and seed sensitivity

Because the event-discovery panel uses ten negatives per positive, we repeated the analysis with independent negative

draws. A fixed linear scorer produced AUROC 0.74320 and 0.74317 on two 1:10 samples, while sampled AUPR was 0.20848 and 0.20847. Changing the ratio to 1:5 increased sampled prevalence from 9.09% to 16.67% and sampled AUPR to 0.34134. Hence AUROC is the safer cross-sample comparison; sampled AUPR and precision@100 must retain the sampling contract.

With seed 7 in addition to seed 42, FT-Transformer first-pair AUPR was 0.14387 and 0.14665 (mean 0.14526, SD 0.00197), while GraphSAGE fusion was 0.15039 and 0.15353 (mean 0.15196, SD 0.00222). Recurrence AUPR changed by less than 0.001 for both. These checks support reproducibility of the ordering but are not a substitute for multi-year replication.

4.4. Interpretation of the negative controls

For triplet recurrence, the strongest event-level signal is historical pair–event count (AUPR 0.85475), whereas the lower 2.5th-percentile bound of the Ω shrinkage measure [2], denoted $\Omega_{0.25}$, reaches 0.68464. This is expected: count predicts repeated reporting, while Ω measures excess reporting against an additive-risk baseline [3] and cannot rank a genuinely unseen triplet without an observed count. These recurrence-based AUPRs are evaluated on a high-prevalence panel and are not comparable to the low-prevalence first-reporting AUPR of Table 2. The distinction is a useful negative control against conflating pharmacovigilance signal detection with future-discovery prediction.

4.5. Fold-wise and reporting-role analysis

Fig. 2 gives the fold-wise recurrence results rather than only the pooled frozen score. NEST-DDI is better on all three rolling transitions, with the smallest gap on the second of the three rolling folds and the largest on the later holdout. The later holdout preserves the ordering after all model choices are frozen. This pattern is consistent with a modest temporal advantage, not a one-fold artifact.

We also stratified the later holdout by the dominant FAERS reporting role of a pair. For suspect–suspect, concomitant–concomitant and suspect–concomitant strata, the NEST minus HGB AUPR differences were 0.00052, 0.00049 and 0.00130, respectively. Some small strata favored HGB, indicating that the gain is not invariant to reporting role. Among 55,890 pairs with a known dominant role, no role-specific model was trained; the analysis is descriptive and uses only cutoff-available labels.

At operational review capacities, the later holdout adds one observed positive at the top 10,000 and 59 at the top 50,000 relative to HGB. A cutoff-dated openFDA label-text audit explicitly mentioned 27 of NEST-DDI’s top 300 pairs versus 31 for HGB; label absence is not a negative. Thus discrimination gains do not imply a larger actionable or label-concordant list.

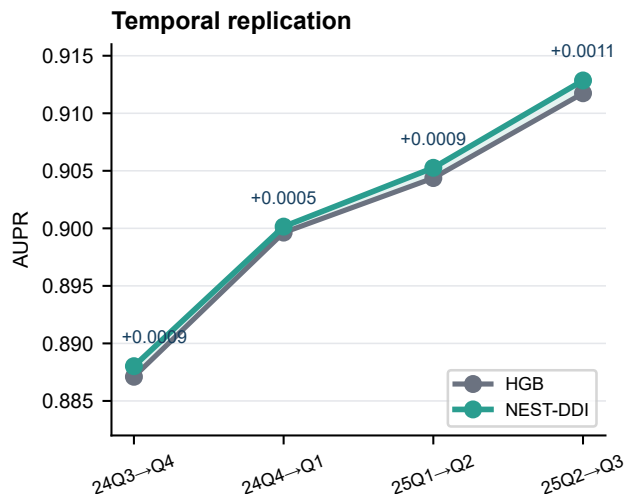


Fig. 2. Fold-wise temporal replication. Annotated gaps are NEST-DDI minus HGB AUPR.

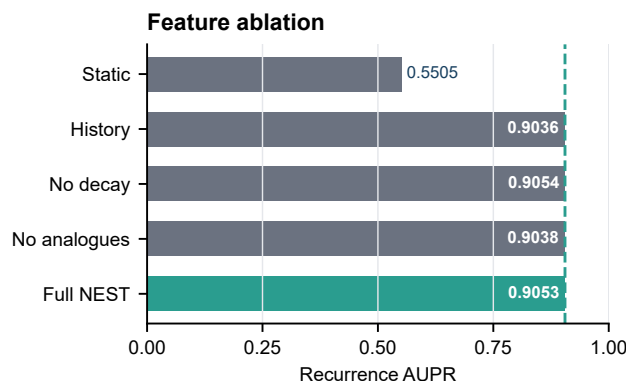


Fig. 3. Feature ablation on the frozen recurrence panel; the dashed line marks full NEST-DDI.

5. LIMITATIONS AND CONCLUSION

FAERS lacks exposure denominators and is affected by indication, publicity, warnings and duplicate submissions; labels therefore represent reporting, not incidence. Drug mapping, the 20-drug cap, five-quarter coverage and case–control sampling limit transportability, while receipt date can differ from exposure or event onset.

This release-local benchmark is not independent clinical validation. Sampled AUPR depends on the negative ratio, and NEST-DDI is operational rather than mechanistic: it does not identify pathways, dose or exposure. Future work should add longer windows, exposure-aware denominators, cold-start splits, versioned mechanism references and preregistered multi-seed review. Separating information time, risk sets and label-opening dates keeps claims auditable; the recurrence gain remains small and the first-pair fusion exploratory.

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7. COMPLIANCE WITH ETHICAL STANDARDS

This study is a retrospective secondary analysis of de-identified, publicly released spontaneous report data from FAERS/AEMS. No human or animal subjects were involved, and ethical approval was not required. The authors have no competing interests to declare.

8. REFERENCES

- [1] U.S. Food and Drug Administration, “FDA Adverse Event Monitoring System (AEMS) latest quarterly data files (formerly FAERS),” 2026. [Online]. Available: [FDA AEMS quarterly data files](#).
- [2] G. N. Norén, R. Sundberg, A. Bate, and I. R. Edwards, “A statistical methodology for drug–drug interaction surveillance,” *Stat. Med.*, vol. 27, no. 16, pp. 3057–3070, 2008.
- [3] B. T. Thakrar, S. B. Grundschober, and L. Doessegger, “Detecting signals of drug–drug interactions in a spontaneous reports database,” *Br. J. Clin. Pharmacol.*, vol. 64, no. 4, pp. 489–495, 2007.
- [4] N. P. Tatonetti, P. P. Ye, R. Daneshjou, and R. B. Altman, “Data-driven prediction of drug effects and interactions,” *Sci. Transl. Med.*, vol. 4, no. 125, p. 125ra31, 2012.
- [5] G. Xiong et al., “DDInter: An online drug–drug interaction database towards improving clinical decision-making and patient safety,” *Nucleic Acids Res.*, vol. 50, no. D1, pp. D1200–D1207, 2022.
- [6] U.S. FDA, “openFDA drug labeling API,” 2025. [Online]. Available: [openFDA API portal](#).
- [7] P. N. Hameed, K. Verspoor, S. Kusljic, and S. Halgamuge, “Positive–unlabeled learning for inferring drug interactions based on heterogeneous attributes,” *BMC Bioinformatics*, vol. 18, no. 1, art. no. 140, 2017.
- [8] J. Y. Ryu, H. U. Kim, and S. Y. Lee, “Deep learning improves prediction of drug–drug and drug–food interactions,” *Proc. Natl. Acad. Sci. USA*, vol. 115, no. 18, pp. E4304–E4311, 2018.
- [9] Y.-H. Feng, S.-W. Zhang, and J.-Y. Shi, “DPDDI: A deep predictor for drug–drug interactions,” *BMC Bioinformatics*, vol. 21, no. 1, art. no. 419, 2020.
- [10] Y. Deng et al., “A multimodal deep learning framework for predicting drug–drug interaction events,” *Bioinformatics*, vol. 36, no. 15, pp. 4316–4322, 2020.
- [11] M. Zitnik, M. Agrawal, and J. Leskovec, “Modeling polypharmacy side effects with graph convolutional networks,” *Bioinformatics*, vol. 34, no. 13, pp. i457–i466, 2018.
- [12] Y. Yu, K. Huang, C. Zhang, L. M. Glass, J. Sun, and C. Xiao, “SumGNN: Multi-typed drug interaction prediction via efficient knowledge graph summarization,” *Bioinformatics*, vol. 37, no. 18, pp. 2988–2995, 2021.
- [13] Z. Xiong et al., “Multi-relational contrastive learning graph neural network for drug–drug interaction event prediction,” *AAAI*, 2023, pp. 5339–5347.
- [14] X. Su et al., “Dual-channel learning framework for drug–drug interaction prediction via relation-aware heterogeneous graph transformer,” *AAAI*, 2024, pp. 249–256.
- [15] D. Xu, C. Ruan, E. Korpeoglu, S. Kumar, and K. Achan, “Inductive representation learning on temporal graphs,” *ICLR*, 2020.
- [16] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein, “Temporal graph networks for deep learning on dynamic graphs,” arXiv:2006.10637, 2020.
- [17] D. S. Wishart et al., “DrugBank 5.0: A major update to the DrugBank database for 2018,” *Nucleic Acids Res.*, vol. 46, no. D1, pp. D1074–D1082, 2018.
- [18] F. Cornell, O. Smirnov, G. Z. Gandler, and L. Cao, “On the power of heuristics in temporal graphs,” *Proc. Mach. Learn. Res.*, vol. 296, 2025, pp. 37–46.
- [19] F. Poursafaei, S. Huang, K. Pelrine, and R. Rabbany, “Towards better evaluation for dynamic link prediction,” *NeurIPS Datasets and Benchmarks Track*, 2022.
- [20] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. ACM SIGKDD*, 2016, pp. 785–794.
- [21] W. L. Hamilton, R. Ying, and J. Leskovec, “Inductive representation learning on large graphs,” in *NeurIPS*, 2017, pp. 1024–1034.
- [22] I. Ramrakhiani, “DRIFT: Drug–drug interaction via foundation transformers,” technical report and code repository, 2026. [Online]. Available: [GitHub](#).
- [23] Y. Chen et al., “DrugDAGT: A dual-attention graph transformer with contrastive learning improves drug–drug interaction prediction,” *BMC Biol.*, vol. 22, no. 1, art. no. 233, 2024.
- [24] R. Kpanou, P. Dallaire, E. Rousseau, and J. Corbeil, “Learning self-supervised molecular representations for drug–drug interaction prediction,” *BMC Bioinformatics*, vol. 25, no. 1, art. no. 47, 2024.
- [25] H. Mei and J. Eisner, “The neural Hawkes process: A neurally self-modulating multivariate point process,” *NeurIPS*, 2017.
- [26] S. Zuo et al., “Transformer Hawkes process,” *ICML*, 2020.
- [27] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *ICML*, 2017, pp. 1321–1330.